# 18-Day .NET, C# & Vue.js Study Checklist

A learn-by-doing roadmap covering C# syntax, Vue 3 fundamentals, ASP.NET Web APIs, and full-stack deployment.

| DONE | DAY | FOCUS TOPIC | ACTIONABLE TASK |
|---|---|---|---|
| **Phase 1: C# Foundations** | | | |
| ☐ | **Day 1** | **C# Setup & First Code** | Set up Visual Studio / VS Code + .NET SDK. Complete freeCodeCamp C# Module 1 (variables, strings, basic math). |
| ☐ | **Day 2** | **Logic & Control Flow** | Learn if/else statements, switch cases, and boolean expressions. Build a simple console decision game. |
| ☐ | **Day 3** | **Loops & Collections** | Master for, foreach, and while loops along with Arrays and Lists. Build a basic console To-Do list. |
| ☐ | **Day 4** | **Methods & Error Handling** | Write reusable methods with parameters/return types and practice try-catch blocks for clean error handling. |
| ☐ | **Day 5** | **Object-Oriented Basics** | Learn classes, objects, properties, and constructors. Model a simple MenuItem class to prepare for the main project. |
| **Phase 2: Vue 3 Foundations** | | | |
| ☐ | **Day 6** | **Vue 3 Setup & Reactivity** | Start Bhrugen Patel's Udemy course. Create your first Vue 3 project and learn ref(), reactive(), and template syntax. |
| ☐ | **Day 7** | **Directives & Event Handling** | Practice v-if, v-for, v-bind, and v-on (@click). Build an interactive dynamic item counter and filterable list. |
| ☐ | **Day 8** | **Components & Props** | Break UI into reusable child components. Pass data down with Props and send events up with Emits. |
| ☐ | **Day 9** | **Forms & Two-Way Binding** | Use v-model to handle text inputs, checkboxes, and select dropdowns with basic form validation. |
| ☐ | **Day 10** | **Vue Router & Navigation** | Set up multi-page navigation using Vue Router, route parameters, and navigation guards. |
| **Phase 3: .NET Backend API** | | | |
| ☐ | **Day 11** | **.NET Web API & Database** | Initialize the ASP.NET Core Web API project. Configure Entity Framework Core and connect to SQL Server / SQLite. |
| ☐ | **Day 12** | **Models, Migrations & CRUD** | Create database models, run EF Core migrations, and build GET, POST, PUT, and DELETE API endpoints. |
| ☐ | **Day 13** | **Authentication & Security** | Implement ASP.NET Identity and JWT token authentication for user registration, login, and role-based access. |

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|---|---|---|---|
| **Phase 4: Full-Stack Build** | | | |
| ☐ | **Day 14** | **Connecting Vue to .NET** | Configure CORS on the .NET API. Use Axios or Fetch in Vue 3 to retrieve and display menu items from your backend. |
| ☐ | **Day 15** | **Admin CRUD Operations** | Build protected admin pages in Vue to create, edit, and delete restaurant items with image uploads. |
| ☐ | **Day 16** | **Shopping Cart & State** | Implement Pinia (or reactive state) to manage the user's shopping cart, item quantities, and order totals. |
| ☐ | **Day 17** | **Search, Filter & Orders** | Add category filtering, keyword search, and complete the end-to-end order checkout flow. |
| ☐ | **Day 18** | **Testing & Cloud Deployment** | Test all user/admin workflows, fix edge-case bugs, and deploy your full-stack app to Azure. |